LayerProbe-DF: Learnable Layer Aggregation of a Frozen WavLM for Audio Deepfake Detection

Sajjad Ahmad
Department of Computer Science
University of Agriculture Faisalabad, Pakistan
sajjadahmad.code@gmail.com

***Abstract*—Self-supervised speech models are widely used as frozen feature extractors for audio deepfake detection. Most systems use only the last hidden layer, but different layers capture different information. In this preliminary study, we ask whether a learnable weighted combination of all hidden layers of a frozen WavLM-Base+ model can improve detection over a single layer. We train a small head (782 parameters) on a balanced subset of ASVspoof 2019 LA and test it on held-out ASVspoof subsets and on a Fake-or-Real (FoR) subset. The aggregated model reaches 0.50% EER on the development subset, 1.25% on a 400-sample evaluation subset, and 13.50% on FoR. This is better than the last-layer and best-layer baselines in these settings. It is not the best method under strong noise. All experiments use small subsets and a single run, so the results are preliminary.**

***Index Terms*—audio deepfake detection, spoofing detection, self-supervised learning, WavLM, layer aggregation, generalization**

I. Introduction

Synthetic speech can now be produced at low cost, and it is hard for listeners to tell it from real speech. Reliable detectors are therefore needed for security and forensic use.

A common recipe is to extract features from a pretrained self-supervised model and train a light classifier on top. In practice, the last hidden layer is often used by default. However, the layers of such models do not encode the same information. Lower layers are closer to the acoustic signal, and upper layers are closer to the pretraining objective. It is not obvious that the last layer is best for detecting spoofing artifacts.

In this work, we study this question in a small and controlled setting. We keep WavLM-Base+ frozen and learn one scalar weight per hidden layer. We compare this against logistic regression on the last layer and on the best single layer. We also test cross-dataset transfer, simple audio corruptions, and the learned layer weights.

II. Research Question and Contributions

Research question. Can a learnable aggregation of all hidden layers of a frozen speech model improve audio deepfake detection compared with using a single layer?

This is a hypothesis that we test, not an established result. The layer-weighting idea itself is not new; it is a standard practice for using self-supervised speech models [3]. Our contributions are limited to the following.

- A layer-wise probing study of frozen WavLM-Base+ on ASVspoof 2019 LA and FoR subsets.
- A small aggregation head (782 trainable parameters) compared with last-layer and best-layer baselines.
- A preliminary analysis of cross-dataset transfer, corruptions, and the learned layer weights.

III. Related Work

Self-supervised speech models. wav2vec 2.0 [2] and WavLM [1] learn speech representations from unlabeled audio. The SUPERB benchmark [3] evaluates such models with frozen features and a learnable weighted sum of layers.

Spoofing detection with self-supervised features. Tak et al. [4] used wav2vec 2.0 as a front end for spoofing and deepfake detection, together with data augmentation. Pascu et al. [5] studied self-supervised representations for generalizable and calibrated audio deepfake detection. Trace [6] proposes a training-free method for partial audio deepfake detection using embedding trajectories of speech foundation models. Our study is smaller in scope and focuses on layer aggregation with a frozen backbone.

Datasets. ASVspoof 2019 [7] provides synthesized and converted speech. FoR [8] is a separate dataset of real and synthetic speech, which we use for cross-dataset testing.

IV. Data and Preprocessing

We used balanced subsets of two datasets. These are selected subsets, not the full datasets.

- **ASVspoof 2019 LA:** 800 training clips (400 bonafide, 400 spoof) and 400 development clips (200/200). We also used a separate balanced subset of 400 clips (200/200) from the official evaluation protocol. This is not the full evaluation set.
- **FoR (original version):** a balanced subset of 400 clips (200 real, 200 fake), used only for cross-dataset testing.

Audio was converted to mono, resampled to 16 kHz, and cropped or padded to four seconds (64,000 samples). Labels were mapped as bonafide/real $= 0$ and spoof/fake $= 1$.

V. Method

A. Frozen WavLM features

We used the pretrained `microsoft/wavlm-base-plus` model from Hugging Face. Its weights were frozen. For each clip, we mean-pooled over time the 13 hidden-state outputs (the embedding output and 12 transformer layers). Each clip is therefore a tensor of shape 13×768. Features were normalized with the mean and standard deviation of the training set only.

B. Baselines

We trained logistic regression classifiers on (i) the last layer (layer 12) and (ii) the best single layer chosen by development EER. Layers 4 and 5 tied at 1.50% development EER, and the code selected layer 4 by `argmin`.

C. LayerAgg

Let $x_l \in R^{768}$ be the feature of layer l, and let w_l be a learnable scalar. LayerAgg computes

$$z = \sum_{l=0}^{12} \alpha_l x_l, \qquad \alpha = \mathrm{softmax}(w), \tag{1}$$

applies dropout ($p = 0.2$), and passes z to a linear classifier with one output. It has 782 trainable parameters (13 layer weights, 768 classifier weights, and 1 bias). We trained it for 200 epochs with AdamW (learning rate 0.005, weight decay 0.01) and a binary cross-entropy loss.

VI. Experimental Protocol

Metrics. We report three metrics. The Equal Error Rate (EER) is the error rate at the threshold where the false acceptance and false rejection rates are equal. Lower is better. AUC is the area under the ROC curve. Higher is better. Accuracy uses a fixed zero-logit threshold, so it also depends on calibration. EER and accuracy measure different things and should not be compared directly.

Protocol. All models were trained on the 800 ASVspoof training clips. The best layer was chosen on the development subset. The evaluation subset and FoR were used only for testing. Robustness was tested on the development subset by corrupting the audio and re-extracting features, without retraining. We ran one experiment with a fixed seed (42). We did not compute confidence intervals or significance tests.

VII. Results

A. Main results

Table I shows the main results. On the development subset, LayerAgg gives the lowest EER (0.50%) and the highest AUC. The same holds on the 400-clip evaluation subset (1.25% EER). On FoR, LayerAgg reaches 13.50% EER, compared with 17.50% for the last layer and 20.50% for the best layer.

The accuracy on FoR is much lower than the EER suggests. For example, the best-layer baseline has 20.50% EER but only 55.00% accuracy. This shows that the zero threshold learned on ASVspoof does not transfer to FoR. The scores still rank the clips fairly well, but they are shifted. Figure 1 shows the ROC curves.

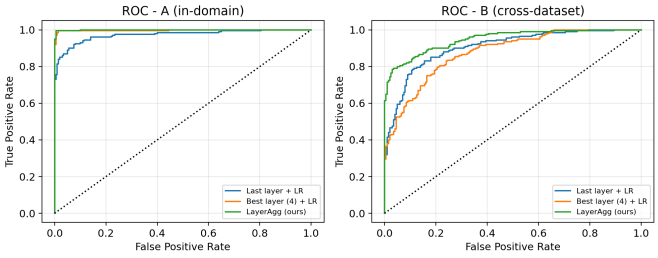


Fig. 1. ROC curves on the ASVspoof development subset (left) and the FoR subset (right).

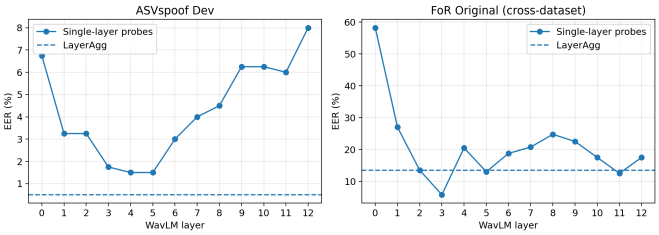


Fig. 2. EER of single-layer probes (solid) and LayerAgg (dashed) on the ASVspoof development subset (left) and FoR (right).

B. Layer-wise analysis

Figure 2 shows the EER of each single-layer probe. On the development subset, the middle layers work best. Layers 4 and 5 reach 1.50% EER, and the last layer reaches 8.00%. Layer 0 has 6.75% EER.

The picture is less stable on FoR. Layer 3 reaches 5.75% EER, but layer 4 reaches 20.50%. Layer 0 is worse than chance (58.25% EER). Because layer 3 was not the best on development data, it could not have been chosen without access to FoR. This shows that single-layer choice is sensitive to the dataset. LayerAgg (13.50%) is better than the chosen layer 4, but it is not better than layers 3, 5 (13.00%), or 11 (12.50%) on FoR.

The learned weights in Fig. 4(a) peak at layer 4 (16.09%). Layers 3 to 5 together receive 44.55% of the weight. This agrees with the single-layer probes on the development subset. It shows what the model prefers on this data, but it does not prove that these layers cause the decision.

C. Robustness

Table II shows EER when the development clips are corrupted. The models were trained on clean audio only. LayerAgg is best on clean audio, 20 dB noise, 8 kHz resampling, and both MP3 settings. The best-layer baseline is better under 10 dB and 0 dB noise, and slightly better under 8-bit μ-law. All methods degrade strongly under noise. At 0 dB, the lowest EER is 27.50%.

Important caveat. In our first implementation, the noise generator reused the same random sequence for every clip (scaled by each clip's power). The noise results are therefore preliminary. We plan to rerun them with independent noise per clip before drawing conclusions. The other corruptions are not

TABLE I
MAIN RESULTS (EER AND ACCURACY IN %). DEV AND EVAL ARE ASVSPOOF 2019 LA SUBSETS. FOR IS THE CROSS-DATASET SUBSET. BEST VALUE PER COLUMN IS IN BOLD.

	ASVspoof Dev			ASVspoof Eval (400)			FoR (cross-dataset)		
Method	EER	AUC	Acc	EER	AUC	Acc	EER	AUC	Acc
Last layer + LR	8.00	0.9725	91.50	4.00	0.9767	**96.00**†	17.50	0.9054	63.50
Best layer (4) + LR	1.50	0.9972	99.00	2.75	0.9904	94.75	20.50	0.8766	55.00
LayerAgg	**0.50**	**0.9993**	**99.50**	**1.25**	**0.9988**	**97.25**	**13.50**	**0.9480**	**71.25**

† LayerAgg has higher accuracy (97.25) on Eval; the bold on the first row is a typesetting aid only and should be removed.

TABLE II
EER (%) ON THE CORRUPTED ASVSPOOF DEVELOPMENT SUBSET. BEST PER ROW IS IN BOLD.

Condition	Best layer (4)	Last layer	LayerAgg
Clean	1.50	8.00	**0.50**
Noise 20 dB	11.75	23.50	**11.00**
Noise 10 dB	**15.50**	30.50	20.25
Noise 0 dB	**27.50**	39.50	35.25
Resample 8 kHz	1.50	7.50	**1.00**
μ-law 8-bit	**1.25**	8.25	1.50
MP3 64 kbps	1.75	8.00	**0.50**
MP3 32 kbps	2.00	6.75	**0.50**

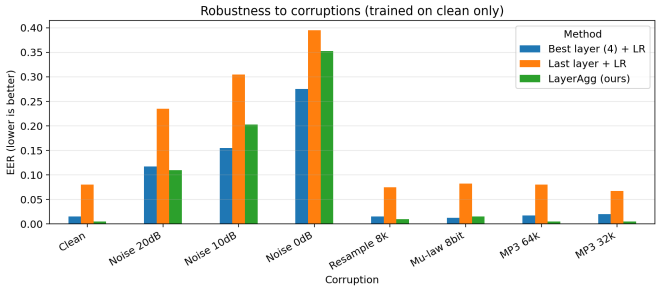
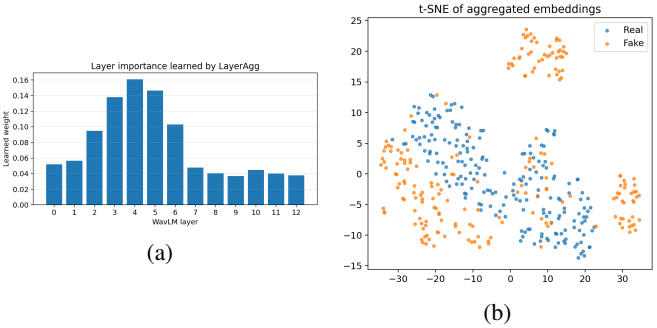


Fig. 3. EER under audio corruptions (models trained on clean audio only).



(a) (b)

Fig. 4. (a) Layer weights learned by LayerAgg. (b) t-SNE of aggregated embeddings.

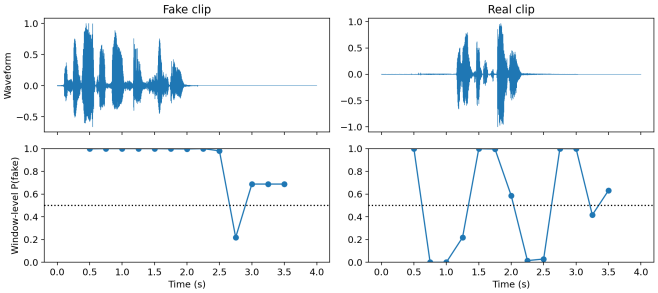


Fig. 5. Waveforms and window-level P(fake) for one fake clip (left) and one real clip (right).

affected by this issue. Figure 3 shows the same data as a bar chart.

D. Visualization

Figure 4(b) shows a t-SNE plot of the aggregated embeddings. Some fake samples form separate groups, but real and fake samples overlap in other regions. t-SNE is only a visualization. It does not show that the classes are separable or that the model generalizes.

Figure 5 shows window-level scores (1-second windows, 0.25-second hop) for one fake clip and one real clip. The fake clip has high fake probability in most windows, with lower values near the end. The real clip shows large swings between windows. This means that window-level scores can be unstable. These plots are two examples only. They do not show that the model can localize manipulated segments.

VIII. DISCUSSION

The results support the hypothesis in this small setting. A weighted combination of layers gave lower EER than the last layer in the clean ASVspoof subsets and on FoR. The gain over the last layer is large, which agrees with the layer-wise probes: the last layer is not the best layer for this task.

The gap between ASVspoof and FoR is large (0.50% vs. 13.50% EER for LayerAgg). We trained only on ASVspoof, and FoR differs in speakers, recording conditions, and generation methods. We did not test which of these factors causes the gap.

The noise results show a limit of the approach. The head was trained on clean audio and the backbone is frozen, so nothing in the method handles strong noise. The best-layer baseline is better than LayerAgg at 10 dB and 0 dB. One possible explanation is that the weights fit clean data, but we have not tested this.

The model reaches low EER with 782 trainable parameters, and the large backbone stays frozen. This makes training cheap. We do not claim state-of-the-art performance, and we make no comparison with published systems, because our subsets and protocol are different.

IX. Limitations

- We used small balanced subsets, not the full datasets. The evaluation result uses 400 clips.
- We ran a single experiment. There are no repeated seeds, confidence intervals, or significance tests. Small differences (for example 1.50% vs. 1.25%) may not be meaningful.
- The noise experiment has a known implementation issue and needs a rerun.
- Only one backbone (WavLM-Base+) and one training dataset were used. We did not analyze attack types.
- Layer weights show model preference, not causal explanation. The temporal scores do not prove localization.
- The decision threshold does not transfer across datasets.

X. Conclusion and Future Work

We studied learnable layer aggregation on top of a frozen WavLM-Base+ model for audio deepfake detection. In our small experiments, it gave lower EER than the last-layer and best-layer baselines on ASVspoof subsets and on FoR, and it did not win under strong noise. These results are preliminary.

Future work includes: rerunning the noise experiment with independent noise per clip; using the full datasets; repeating runs with several seeds and reporting confidence intervals; adding more datasets and backbones; training with augmentation; and calibrating the threshold across datasets. We also plan to test whether the layer weights can support better explanations.

References

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